SGA 2: Application to the Fundamental Group

Purity theorem — proof of part (ii)

Bounded source-aligned working unit — 22 July 2026

Authority and boundary. Corrected French TeX lines 3573–3574: the complete proof of part (ii) and its proof close. The text is on original printed p. 122, source-PDF physical p. 105, recomposed running p. 97. Blank line 3575 is the raw continuation cursor; line 3576 (Theorem 3.10) is the next substantive cursor. French remains byte-identical. Same-lineage source PDFs are manifestation evidence only; the external English candidate is comparison only. This no-overwrite successor is internal and pending fresh independent review.

Consolidated source note. French line 3573 has three adjudicated defects: "Pouvons" must express the proof transition "Let us prove" [SGA2-X-L3573-POUVONS-VS-PROUVONS-SRCDEF-001]; the completion reduction must cite Corollary 3.8, not the nonexistent "Corollary 3.9" [SGA2-X-L3573-COR39-VS-COR38-XREF-SRCDEF-001]; and the grammatical phrase "un intersection complète" is rendered "a complete intersection" [SGA2-X-L3573-UN-VS-UNE-INTERSECTION-SRCDEF-001]. The French authority has not been patched.

Let us prove (ii). Let A be a noetherian local ring of dimension at least 3. Suppose that there exist a regular noetherian local ring B and a B-sequence $(t_1, \dots, t_k)$ such that

$$A \simeq B/(t_1, \dots, t_k).$$

We prove that A is pure by induction on k. If $k = 0$, this follows from (i). Suppose that $k \geq 1$ and that the result has been proved for $k' < k$. By Corollary 3.8, we may assume that A, and hence also B, is complete. Set

$$C = B/(t_1, \dots, t_{k-1}).$$

Then $A \simeq C/t_k C$, and t_k is C-regular. By the induction hypothesis, C is pure; it therefore suffices to verify that Lemma 3.9(ii) is applicable. In the notation of that lemma, its A and B are here C and A, respectively. We have $\dim C \geq 4$; hence, for every prime ideal $\mathfrak{p}$ of C such that $\dim(C/\mathfrak{p}) = 1$, one has $\operatorname{depth} C_{\mathfrak{p}} \geq 3$. Moreover, $C_{\mathfrak{p}}$ is a complete intersection with $k' \leq k - 1$, and is therefore pure by the induction hypothesis.